

Maxwell DeFanti

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EDUCATION

University of California, Berkeley

Aug. 2024 – May 2028

Bachelor of Arts in Applied Mathematics + Data Science

Berkeley, CA

- Machine Learning, Probability and Random Processes, Convex Optimization, Abstract Algebra, Linear Algebra, Differential Equations, | AI Interpretability: Meanings, Limits, and Methods, AI Safety and Alignment

EXPERIENCE

Deep Learning Researcher

September 2025 – Present

Berkeley Artificial Intelligence Research

Berkeley, CA

- Designed and built novel Continual Compress Refresh algorithm with **PyTorch** that **completely eliminates loss of plasticity** in standard Feed-Forward, Convolutional, and Residual **neural networks** while outperforming the previous state of the art with an **over 9% reduction in online test loss** on OpenAI Gymnasium Reinforcement Learning problems
- Built upon existing statistical theory of neural network plasticity loss under L2 regularization to develop novel **Gaussian Layer-Norm Rescaling** method, which periodically rescales neurons' ingoing and outgoing weight norms with dynamically parameterized Gaussian Noise in order to prevent functional collapse. GLNR improves both standard backpropagation and continual backpropagation network performance in continual learning settings by **11-20%** across multiple loss metrics

Machine Learning Engineering Intern

May 2026 – August 2026

Capital One

Richmond, VA

- Developed **AI agents** to intelligently query **50+ petabytes** of **AWS log data** for ML applications, removing manual overhead, resulting in a **24x efficiency increase** as measured by Databricks billing, and allowing for instant response time.
- Automated and standardized the integration, metadata enrichment, and resource provisioning of new data sources into Capital One's internal AWS log data pipeline, eliminating repetitive engineering work and organization level downtime

Artificial Intelligence Safety Research Fellow

June 2026 – September 2026

Cambridge University

Hybrid

- Selected as one of **75** research fellows out of over **1750** applicants for Cambridge AI Safety Hub's MARS program
- Fine-tuned LLMs** to enforce Bayesian belief updates via the **Martingale Property** using Low-Rank Adaptation (**LoRA**), improving forecasting performance **Brier score: 0.29 → 0.23** and led the design and implementation of simulated human AI interaction environments in order to scan for and mitigate **AI sycophancy** and **belief entrenchment**

Software Engineering Intern

June 2025 – May 2026

The Materials Project

Lawrence Berkeley National Lab, Berkeley, CA

- Built critical **full-stack applications** for admin side **database management**, filtration, and modification using **dash.py** and the **MPContribs** api for a DOE funded research initiative with **30,000+ citations**
- Developed a novel web interface that enabled **600,000+** users of the **Materials Project** to upload and modify their research data, implementing custom **React** components, efficient information extraction across file types and flexible data formatting with **Pandas**, and seamless **API** integration

PUBLICATIONS

Towards Unsupervised Martingale Training

DeFanti, et al.

NeurIPS 2026 Workshop: Dynamic Alignment in Human-AI Coupled Systems

under review

PROJECTS

Activation Probes for Synthetic Toxin Variant Detection | *Python, PyTorch, ESM-2, Scikit-Learn*

April 2026

- Trained a **two-layer MLP activation probe** on frozen **ESM-2 650M** mean-pooled embeddings of natural and synthetic toxins using weighted binary cross-entropy, **AdamW**, and **CUDA**-accelerated training and inference
- Achieved **86.7% recall** on synthetic toxin variants at ~40% sequence identity where BLAST collapsed to **46.7%**, with an overall **AUC-ROC of 0.983**

Multilevel LLM Jailbreak Security | *Python, PyTorch, Huggingface Transformers*

May 2026

- Implemented **Greedy Coordinate Gradient** suffixes as well as other common jailbreak attack strategies in a testing environment against a **2.5 billion parameter Qwen model AdamW**, and **CUDA**-accelerated training and inference
- Leveraged **Natural Language Processing** methods to achieve **87.0% improvement** in jailbreak refusal rate over the base model identity with a **custom multilevel defense algorithm** that takes advantage of **self reminding**, **perplexity filtration**, and **LLM judging**

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML/CSS, MATLAB, SQL, Linux

Libraries: PyTorch, Scikit-learn, Cuda, Pandas, NumPy, Huggingface Transformers, Matplotlib, OpenCV, MATLAB Engine, NLTK, LangChain, MPContribs, PyMOL